

OMIS 105

Welcome Notebook & Tech Stack

Quarter: Fall 2026

Instructor: Dr. Mahmoud Parsian (mparsian@scu.edu)

Tech Stack

- **Language:** Python · SQL
- **Database:** DuckDB (in-memory)
- **Notebooks:** Marimo (reactive)
- **Audience:** Junior/Senior business students
- Zero prior exposure to notebooks, SQL, or databases

Purpose

Day-one onboarding notebook for
OMIS 105 — Introduction to Database Management Systems

This is the first thing students open.

- Introduces Marimo
- Introduces the concept of a database
- Walks through the very first SQL queries

👉 One self-guided, interactive notebook

Files in This Folder

File	Purpose
SQL_101_DuckDB_Notebook_1.py	Day-one student notebook
SQL_101_DuckDB_Notebook_2.py	Day-one student notebook
Outline_OMIS_105_10_weeks.md	10-week course outline

Notebook Structure (1/2)

1. **What is a notebook?** — Cells, text vs SQL, Cmd/Ctrl+Enter
2. **What is a database?** — Tables = spreadsheets with rows/columns
3. **First table:** `students` — CREATE TABLE + INSERT, 6 rows
4. **Asking questions with SQL** — three business questions:
 - "Who are the Marketing majors?" → WHERE
 - "Who likes Pizza?" → WHERE (different column)
 - "How many students in each major?" → GROUP BY + COUNT

Notebook Structure (2/2)

- 5. **Try It Yourself** — editable SQL cell, guided suggestions
- 6. **Why Marimo is reactive** — automatic cell updates explained
- 7. **Course roadmap** — 10-week overview table

Marimo Conventions (Pure SQL Cells)

- SQL cells use `_df = mo.sql(f"""...""")` with a bare `return`
- `duckdb.connect(database=':memory:')` created, NOT returned (Marimo auto-discovers the connection)
- Markdown cells use `mo.md("""...""")` with `hide_code=True`
- Use `--` SQL comments, not Python `#`, inside SQL cells
- `CREATE OR REPLACE TABLE` for re-runability

Design Decisions (1/2)

- **Favorite foods, not business data.**

Low-stakes data (Pizza, Sushi, Tacos) keeps focus on the tool, not the business scenario. Business data starts in Week 1.

- **Only 6 rows.** Small enough to see everything at a glance.
- **Three queries only.** Enough to show the pattern —
SELECT + FROM + WHERE, then GROUP BY.

Design Decisions (2/2)

- **"Try It Yourself" cell.**

Hands-on editing builds comfort with the tool.

Specific suggestions lower the barrier.

- **No plots.** Charts come in Week 3.

Day one is about reading tables and writing SQL.

Teaching Notes

- **In-class usage:** Open live. Walk through the first few cells together (5 min), then give students 10 minutes to edit "Try It Yourself" on their own laptops.
- **Reactivity demo:** After editing, scroll back up to show other cells didn't break. "Marimo keeps everything consistent."
- **Common question:** "Where is the data stored?"
→ In memory only. Data disappears when the notebook closes.

Related Materials

Folder	Content
<code>weekly_reviews/</code>	Weeks 1–3, 4–6 notebooks, CSVs, plot helpers
<code>software_installation/</code>	Install guides, setup script, verification
<code>data_stories/</code>	Standalone Python + DuckDB CRUD demos

Let's Get Started

OMIS 105 — Introduction to Database Management Systems — Fall 2026